AP PRECALCULUS

Unit 1 • Worksheet 1.7-1.11B • Answer key

Review 1.7-1.11B • Review of Topics 1.7 to 1.11

Answer key with worked solutions

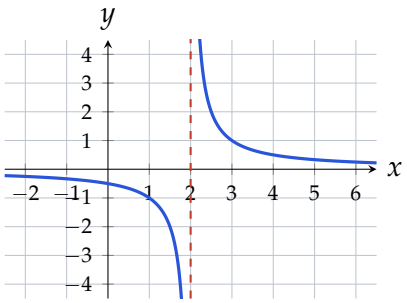
Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1 (1.9 • graph)

NO CALCULATOR

The graph of a rational function s has a vertical asymptote $x = 2$, horizontal asymptote $y = 0$, and passes through $(3, 1)$ and $(1, -1)$. Which could be $s(x)$?



(A) $\frac{1}{x+2}$

(B) $\frac{1}{x-2}$

(C) $\frac{x}{x-2}$

(D) $\frac{x-2}{x}$

Solution

Answer: (B). Asymptote $x = 2$, $y = 0$; $s(3) = 1$.

Question 2 (1.11)

NO CALCULATOR

$\lim_{x \to -3} \frac{x^2 + x - 6}{x^2 - 9}$ is

(A) $\frac{5}{6}$

(B) 0

(C) $-\frac{5}{6}$

(D) It does not exist.

Solution

Answer: (A). $\frac{x-2}{x-3} \to \frac{-5}{-6}$.

Question 3 (1.7) NO CALCULATOR

For large positive x, $s(x) = \dfrac{x^2+x-1}{x-1}$ is closest to which line?

(A) $y = x$
(B) $y = x+1$
(C) $y = x+2$
(D) $y = 1$

Solution
Answer: (C). $s = x+2+\frac{1}{x-1}$.

Question 4 (Free response • AP style) NO CALCULATOR

Let $r(x) = \dfrac{ax^2+bx+6}{x^2-2x-3}$, where a and b are constants. The graph of r has a horizontal asymptote $y = 2$ and an x-intercept at $(1,0)$.

(A) Find a and b.
(B) Show that the graph has a hole and find its coordinates.
(C) Find the vertical asymptote and solve $r(x) < 0$.

Solution
(A) $a = 2$; $2+b+6 = 0$, $b = -8$.
(B) $r = \frac{2(x-1)(x-3)}{(x-3)(x+1)}$: hole at $x = 3$, height $\frac{4}{4} = 1$: $(3,1)$.
(C) $x = -1$. $\frac{2(x-1)}{x+1} < 0$ on $(-1,1)$.